B.Tech (1st Semester) **GROUP-A** **Batch (2026-2030)**

Course Code-100104 **Engineering Physics** **3 0 0 3**

Course Outcomes (COs)	Course Outcome Statements
CO1	Apply the concepts of wave optics phenomena in various engineering fields.
CO2	Apply the concepts of laser operation and fiber optic propagation in engineering applications
CO3	Apply the concepts of electromagnetics in engineering applications.
CO4	Utilize semiconductor devices, based on their I-V characteristics, for various engineering applications.
CO5	Apply the principles of Quantum mechanics to analyze quantum phenomena in engineering systems.

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
CO1	*TSO 1a.* Explain the phenomenon of interference of light by division of amplitude of light waves. *TSO 1b.* Find the wavelength of a given monochromatic light source using an interference experiment. *TSO 1c.* Find the wavelength of a given light source by using an n-slit Fraunhofer diffraction pattern. *TSO 1d.* Determine the resolving power of a given optical instrument by using the Rayleigh criteria *TSO 1e.* Analyze the polarization of a given light source.	**Unit- 1.0: Wave Optics (8hrs)** 1.1 Interference, Division of amplitude, Newton's Ring experiment, Michelson interferometer. 1.2 Diffraction, Fraunhofer diffraction, Single and double slit, & Circular aperture 1.3 Diffraction, Grating, 1.4 Rayleigh criterion, resolving power of telescope. 1.5 Polarization, double-refracting crystal, Nicol prism.
CO2	*TSO 2a.* Differentiate between ordinary and laser light sources. *TSO 2b.* Derive the relations between Einstein A & B coefficients. *TSO 2c.* Explain the construction and working of the given laser with application. *TSO 2d.* Explain the Propagation mechanism of light in optical fiber and its application. *TSO 2e.* Differentiate between different	**Unit- 2.0: Lasers and Optical Fiber (8hrs)** 2.1 Characteristics of laser light, Einstein's A & B coefficients, Population inversion, pumping mechanism, Optical resonator 2.2 Ruby laser, He-Ne laser, and Semiconductor laser 2.3 Structure of optical fibre, Basic principles (TIR, Acceptance angle and Numerical Aperture), Types of optical fibres. 2.4 Propagation mechanism of light in fibre,

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
	types of optical fiber *TSO 2f.* Explain the use of Lasers and Fiber optic Sensors in communications, medical, defense, and Industries	Attenuation in optical fibres. 2.5 Applications of optical fibres, Fibre optic sensors
CO3	*TSO 3a.* Interpret the role of dielectric constant in modifying the electric field inside a material. *TSO 3b.* Apply Gauss's Law in dielectric media to solve problems involving electric field in capacitors. *TSO 3c.* Classify magnetic materials as dia, para, and ferromagnetic based on their response to an external magnetic field *TSO 3d.* Explain Maxwell's equations in integral form with its physical significance. *TSO 3e.* Calculate the magnitude and direction of Poynting vector for a given EM wave and interpret energy transport per unit area.	**Unit- 3.0**: **Electromagnetism & Propagation of Waves (8hrs)** 3.1 Electric fields, Gauss' Law, Dielectrics and Capacitors. 3.2 Magnetic fields & Magnetic Materials. 3.3 Maxwell's Equations. 3.4 Electromagnetic waves & Poynting Theorem
CO4	*TSO 4a.* Explain the need and requirement for Quantum Mechanics *TSO 4b.* Implement the idea of the De-Broglie hypothesis and uncertainty principle for solving problems *TSO 4c.* Explain the concept of wave functions with physical significance. *TSO 4d.* Solve the fundamental governing equations of quantum mechanics to determine the evolution of quantum systems. *TSO 4e.* Calculate the quantized energy levels and normalized wave functions for a particle confined in a one-dimensional infinite potential well. *TSO 4f.* Analyze the phenomenon of quantum tunneling to explain	**Unit- 4.0: Quantum Mechanics (8hrs)** 4.1 Introduction to Quantum Mechanics. 4.2 De-Broglie hypothesis, Heisenberg uncertainty principle 4.3 Wave function and its characteristics 4.4 Schrödinger's equation (time-dependent and independent) 4.5 Particle in a box, Energy Eigen values and Eigen functions 4.6 Potential barrier and Tunneling

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
	how particles penetrate classically forbidden energy barriers	
CO5	*TSO 5a.* Compare the band structures of metals, insulators and semiconductors to explain the differences in their electrical behavior. *TSO 5b.* Distinguish between drift and diffusion current density to solve problems involving carrier transport in semiconductors. *TSO 5c.* Analyse the significance of Hall coefficient in identifying n-type and p-type semiconductors. *TSO 5d.* Evaluate the suitability of solar cells as renewable energy conversion devices in engineering systems. *TSO 5e.* Explain the properties of nanomaterials with applications. *TSO 5f.* Classify various nanomaterials on the basis of confinement.	**Unit- 5.0: Semiconductors & Nano-materials (10hrs)** 5.1 Introduction, Types of materials (metal, semiconductor and insulator) 5.2 Intrinsic and extrinsic semiconductors, P-type and N-type semiconductors, Diffusion, Drift and Fermi level 5.3 Photodiode, P-N junction transistor, LED 5.4 Hall effect 5.5 Solar cell and its characteristics. 5.6 Introduction to nanoscience& nanotechnology, the significance of nanoscale 5.7 Unique Properties: Comparison of bulk and nanomaterials, large surface-to-volume ratio, change in band gap, and optical/electrical/mechanical properties at the nanoscale. 5.8 Classification of Nanostructured Materials: 0D (Quantum Dot), 1D (Nanowire

Reference/Books

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
1.	Optics	Ajay Ghatak	Mcgraw Hill Education,2012, 5th Edition, ISBN-10: 1259004340
2.	Principles of Lasers	O.svelto	Springer Science &Business Media,2010, 5th Ed, ISBN-10: 1441913017
3.	Introduction to electrodynamics	David J. Griffiths	Pearson Education Cambridge University Press, 2023, 5th Edition, ISBN-10: 1009397729
4.	Principles of Electronic Materials and Devices	S.O. Kasap	McGraw Hill Education, Third Edition, 2006, ISBN-10: 0073104647
5.	Semiconductor Physics and Devices	Donald A. Neuman	McGraw-Hill Education, Fourth Edition, 2011, ISBN-10: 0073529583
6.	Introduction to Quantum Mechanics	David J. Griffiths	Pearson Education Cambridge University Press, 2016, 2nd Edition, ISBN-10: 1107179866

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
7.	Nanotechnology: Principles and Practices.	Sulabha K. Kulkarni	Springer Nature, Third Edition,2014, ISBN-10: 3319091700
8.	Concepts of Engineering Physics	S.K. Roy	Novelty Publication, 1st Edition, 2016-17
9.	Concepts of Engineering Physics	Kumari Mamta and Raj Kumar Singh	CRC Press-Taylor and Francis, United Kingdom, 1st Edition, ISBN 9781041191834
10.	Physics Practical for Engineers	S.K. Roy	Novelty Publication, 1st Edition, 2018